

2.4.9. Prop.

(1)

Proof.

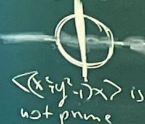
$$\langle F \rangle = \langle \underbrace{g_1, \dots, g_m}_{\text{Gröbner basis of this ideal}} \rangle$$

$g_i \in \langle F_i \rangle$ for some finite subset F_i of F .

$$\Rightarrow g_1, \dots, g_m \in \langle F_1 \cup \dots \cup F_m \rangle \Rightarrow \underbrace{\langle g_1, \dots, g_m \rangle}_{\langle F \rangle} \subseteq \langle F_1 \cup \dots \cup F_m \rangle \Rightarrow \langle F_1 \cup \dots \cup F_m \rangle = \langle F \rangle$$

So, we can fix $f_1, \dots, f_s$ with $\{f_1, \dots, f_s\} = F_1 \cup \dots \cup F_m$. □

$$(x^2 + y^2 - 1) \cdot x = 0$$

 $\langle (x^2 + y^2 - 1)x \rangle$ is not prime.

2.4.9' Example

Consider the curve $C = \{ (t, t^2, t^3) : t \in \mathbb{R} \} \subseteq \mathbb{R}^3$.
The respective ideal is $I(C) = \{ f \in \mathbb{R}[x, y, z] : f \text{ is 0 on } C \}$
By Prop 2.4.9, one can generate $I(C)$ by finitely many polynomials.

$$I(C) = \langle x^2 - y, x^3 - z \rangle$$

$\mathbb{Z}$

all ideals are of the form $f\mathbb{Z}$ with $f \in \mathbb{Z}$
 (every ideal is generated by one element)

(2)

 $k[x]$

every ideal is generated by just one element

DISCUSSION

 $k[x, y]$ $k[x, y, z]$ $\vdots$ $k[x_1, \dots, x_n]$

ideals are finitely generated, but

$$\langle 0 \rangle \subsetneq \langle x \rangle \subsetneq \langle x, y \rangle \subsetneq k[x, y]$$

$$\langle 2 \rangle = \langle 6, 4 \rangle \subseteq \mathbb{Z}$$

$$1 \cdot 6 + (-1) \cdot 4 = 2$$

Def. an ideal $I \subseteq R$ is called prime iff $I \neq R$ and for all $f, g \in R$ one has:
 $f, g \in I \Rightarrow f \in I$ or $g \in I$.

The dimension of R is the maximal length of a chain $I_1 \subsetneq I_2 \subsetneq \dots \subsetneq I_n$
 of prime ideals.

2.5 Properties of Gröbner bases.

2.5.1. Def. For $\alpha = (\alpha_1, \dots, \alpha_n), \beta = (\beta_1, \dots, \beta_n) \in \mathbb{Z}_{\geq 0}^n$

we introduce

$$\alpha \vee \beta = (\max\{\alpha_1, \beta_1\}, \dots, \max\{\alpha_n, \beta_n\}),$$

$$\alpha \wedge \beta = (\min\{\alpha_1, \beta_1\}, \dots, \min\{\alpha_n, \beta_n\}).$$

($\alpha \vee \beta$ is the upper bound and $\alpha \wedge \beta$ is the lower bound on α and β).

2.5.2. Def. Let $f, g \in k[x_1, \dots, x_n] \setminus \{0\}$ with $\text{LM}(f) = x^\alpha$ and $\text{LM}(g) = x^\beta$ (i.e., $\alpha = \text{mdeg } f, \beta = \text{mdeg } g$). We define the S-polynomial of f and g as

$$S(f, g) = \frac{x^{\alpha \vee \beta}}{\text{LT}(f)} f - \frac{x^{\alpha \vee \beta}}{\text{LT}(g)} g$$

$$f = \sum c_\gamma x^\gamma$$

